

Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science

Proposal for Thesis Research in Partial Fulfillment
of the Requirements for the Degree of
Master of Science

Title: Evaluating the Impact of Neuromuscular Massage Interventions on Injury Outcomes
in Professional Football Using Clustering and Synthetic Control Methods

Submitted by: Ryan Hourican
299 Vassar Street
Cambridge, MA 02139

Signature of author: _____

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Laboratory: MIT Sports Lab

Brief Statement of the Problem:

The proposal aims to determine whether a targeted neuromuscular massage intervention reportedly used by Tom Brady and other New England Patriots players has a statistically significant effect on injury outcomes, by comparing treated players to counterfactual injury trajectories constructed from clustered peer groups.

Supervision Agreement:

The program outlined in this proposal is adequate for a Master's thesis. The supplies and facilities required are available, and I am willing to supervise the research and evaluate the thesis report.

Anette Hosoi, Thesis Advisor

Evaluating the Impact of Neuromuscular Massage Interventions on Injury Outcomes in Professional Football Using Clustering and Synthetic Control Methods

Ryan Hourican
Massachusetts Institute of Technology
ryanh24@mit.edu

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Abstract

Professional football players experience high injury risk due to continuous high-impact contact, yet some athletes maintain unusually long and relatively injury-free careers. Tom Brady, one of the most durable quarterbacks in NFL history, has publicly attributed his longevity to a specific neuromuscular massage performed by his personal trainer. Other New England Patriots players reportedly received similar treatment, raising the question of whether this intervention had measurable effects on injury outcomes.

This thesis proposes a data-driven framework to evaluate whether players who received this treatment exhibited statistically different injury patterns relative to comparable athletes who did not. The methodology combines player clustering based on performance and workload characteristics with synthetic control methods adapted from public health applications. By constructing counterfactual “synthetic players,” we estimate the injury outcomes these athletes might have experienced without the treatment.

If successful, this work produces a replicable technique for evaluating individualized performance and recovery claims using public data, contributing to evidence-based sports medicine and injury prevention analytics.

1 Introduction

In professional sports, marginal differences in physical maintenance, recovery techniques, and training protocols can meaningfully influence player performance and longevity. The National Football League (NFL) is particularly injury-prone due to frequent collisions and repetitive high-impact motion. Yet amid this environment, Tom Brady maintained one of the longest quarterback careers in history, missing remarkably few games due to injury. Brady and his trainer have attributed part of this durability to a specific neuromuscular massage treatment, and several New England Patriots teammates reportedly received the same treatment.

Evaluating such claims matters for athlete health and public sports science. If effective, similar methods might be broadly applied or clinically studied. If ineffective, promoting them without validation risks misleading athletes, teams, and sports organizations. However, scientific evaluation

of individualized interventions in professional sports is challenging: randomized trials are infeasible, data are limited, and injury risk is multifactorial.

This proposal outlines a statistical methodology to test whether players receiving this massage intervention exhibited measurably different injury outcomes. By constructing peer comparison groups through clustering and applying synthetic control methods from public health research, we estimate counterfactual injury trajectories and evaluate the treatment’s impact. In particular, we adapt a synthetic control framework recently applied to NFL policies in a public health context by García Bulle et al. [1].

2 Related Work

2.1 Sports Injury Modeling

Sports analytics research has examined correlations between injury likelihood and workload, snap counts, positional stress, and age. These models identify strong trends in risk but typically do not evaluate specific medical or training interventions, largely due to a lack of counterfactual data and ethical constraints around experimental manipulation in professional sports environments. As a result, claims regarding individualized treatment efficacy remain predominantly anecdotal.

2.2 Synthetic Control Methods

Synthetic control methods were originally developed for causal inference in economics and policy evaluation, where randomized experiments are infeasible and observational data must be used to approximate counterfactuals. The method creates a “synthetic unit” constructed from weighted combinations of comparable entities, enabling estimation of outcomes that would have occurred in the absence of an intervention. García Bulle et al. [1] recently applied this approach to the NFL domain to evaluate the epidemiological effects of stadium reopening policies during the COVID-19 pandemic. Their work demonstrates the suitability of synthetic control for data-driven causal inference in football contexts, motivating its adaptation to player-level injury outcomes in this proposal.

2.3 Individualized Treatment Claims in Sports

Although massage therapies, physiotherapy, neuromuscular recovery techniques, and personalized conditioning programs are widely marketed in sports, data-driven evaluation of their effectiveness is rare. Existing studies on recovery interventions focus primarily on small samples or laboratory settings, limiting applicability to elite athletes. No prior work has rigorously analyzed injury outcomes for specific professional players receiving a targeted treatment, motivating the need for the framework proposed here.

3 Proposed Work

This thesis proposes a two-stage statistical framework to evaluate whether a specific neuromuscular massage intervention reduced injury risk among NFL players. Our approach combines (1) cluster-based peer group formation and (2) player-level synthetic control modeling.

3.1 Player Clustering

To establish fair comparison groups, we construct statistical player profiles using multi-season performance and workload measures. Features include:

- offensive, defensive, and special teams snap counts,
- position and position subgroup,
- workload and exposure metrics (e.g., rush attempts, targets, pass attempts, sacks taken),
- known proxies for collision intensity and injury risk.

We will evaluate clustering algorithms, including k -means, Gaussian mixture models, and hierarchical clustering, selecting an approach based on silhouette scores and cluster interpretability. These clusters define peer groups from which synthetic controls will be constructed.

3.2 Identification of Treated Players

Players reported to have received the neuromuscular massage treatment will be identified using media reports, interviews, and trainer disclosures over the relevant seasons. While exact treatment timing may not be fully observable, sensitivity analysis will be used to test robustness to uncertainty in treatment periods.

3.3 Synthetic Control Injury Modeling

For each treated player, we estimate a counterfactual injury trajectory by constructing a weighted “synthetic player” from the most similar cluster members. Our implementation adapts the estimation and weighting framework used by García Bulle et al. [1], but applies it to injury outcomes instead of epidemiological metrics. We define injury outcomes using:

- games missed due to injury,
- probability of injury per snap,
- severity indicators when available from public reports.

The resulting counterfactuals provide an estimate of what the treated player’s injury history might have been in the absence of the massage intervention. Statistical significance will be assessed through placebo tests, permutation analyses, and sensitivity checks against cluster variability.

3.4 Anticipated Challenges

Challenges include incomplete public documentation of treatment exposure, variability in injury reporting practices, and limited sample size. We will address these issues using robustness checks, outcome standardization, and multiple cluster-based control groups to bound estimation uncertainty.

3.5 Preliminary Results

To assess feasibility of our approach, we performed initial clustering of multi-season workload and performance statistics. Results indicate clear feature-driven separation across player types, supporting the use of cluster-conditioned comparison groups for constructing synthetic controls.

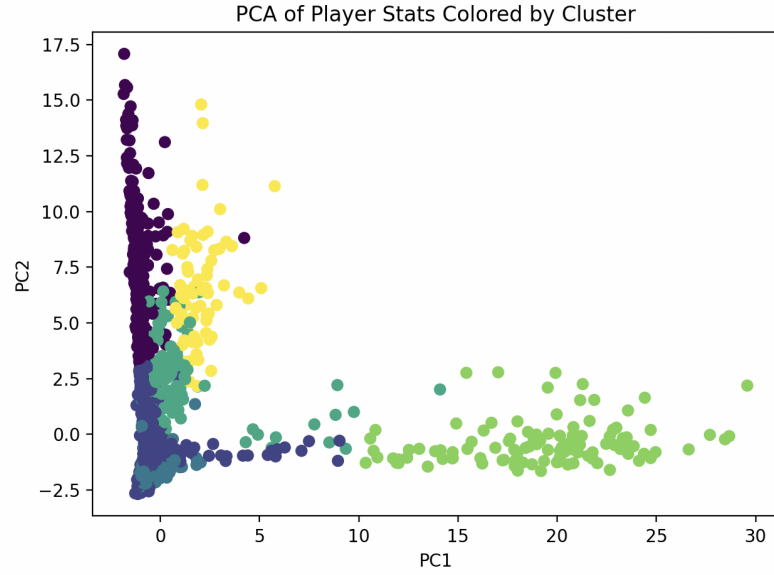


Figure 1: PCA visualization of player statistics colored by cluster, demonstrating workload-based separation.

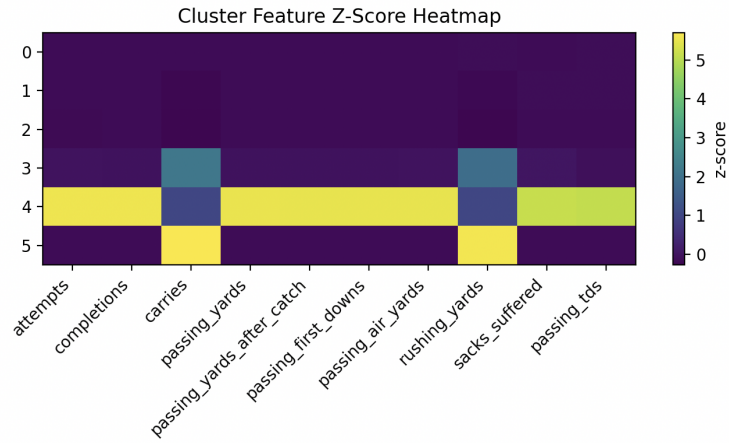


Figure 2: Z-score heatmap of key features by cluster, illustrating differences in workload and usage patterns.

4 Timeline

- **Jan–Feb 2026:** Finalize feature engineering, complete clustering model selection, and identify treated players.
- **Mar 2026:** Implement synthetic control estimator and produce initial counterfactual injury results.
- **Apr 2026:** Conduct robustness checks, placebo tests, and statistical error analysis; finalize quantitative findings.
- **Early May 2026:** Write full thesis document, integrate figures/results, and incorporate advisor revisions.
- **Mid May 2026:** Submit final thesis.

References

- [1] B. García Bulle, D. Shen, D. Shah, and A. E. Hosoi, “Public health implications of opening national football league stadiums during the covid-19 pandemic,” *Proceedings of the National Academy of Sciences*, vol. 119, no. 14, p. e2114226119, 2022.